HINTS 5 Cycle 4 History Document

April 2024

Data Editing

The following variables were identified to contain invalid or unusual values. Those values were replaced with negative value of -4, "Unreadable or Non-conforming numeric response", negative value of -9, "Missing data (Not Ascertained)" or reasonable regular values.

Height_Inches: H6. About how tall are you without shoes? Inches:

One hundred and fifty-four respondents had inches of -9, "Missing data (Not Ascertained)", or -4, "Unreadable or Non-conforming numeric response", which were replaced by 0. The variable Height_Feet is recommended for determining validity of full height.

R_HHAdults: Reconciled number of adults in household

Five respondents had missing values (.), which were replaced by -9.

Totalhousehold: P12. Including yourself, how many people live in your household?

Fifty-eight respondents had Totalhousehold of 0, which were replaced by 1.

DrinksPerDay: J4. During the past 30 days, on the days when you drank, about how many drinks did you drink on average?

One respondent had an outlying value of 50 for DrinksPerDay, which was replaced by -4.

AverageTimeSitting: K4. During the past 7 days, how much time did you spend sitting on a typical day at home or at work?

Fourty-eight respondents had an outlying value of 21 hours or greater for AverageTimeSitting, which was replaced by -4.

The following variables were converted from numeric to character to allow for a leading zero in the variable values: ZIPCODE, ZIP4, FIPS, and FIPST.

Variable Removal

The variables HIGHSPANLI, HISPSURNAME, HISP_HH, PersonID, and QDISP were dropped from the data.

Standard Recode

Standard recode/derived variables are listed below.

AgeGrpA: 4 Level Age Categories Version A (Derived from Age)

The Age variable was re-coded into 4 categories: 18-34; 35-39; 40-44; 45+. The original negative values were carried over.

AgeGrpB: 5 Level Age Categories Version B (Derived from Age)

The Age variable was re-coded into 5 categories: 18-34; 35-49; 50-64; 65-74; 75+. The original negative values were carried over.

EducA: What is the highest level of school you completed? 4 Levels (Derived from Education)

The Education variable was re-coded into 4 categories: Less than High School; High School Graduate; Some College; College Graduate or More. The original negative values were carried over.

EducB: What is the highest level of school you completed? 5 Levels (Derived from Education)

The Education variable was re-coded into 5 categories: Less than High School; High School Graduate; Some College; Bachelor's Degree; Post-Baccalaureate Degree. The original negative values were carried over.

RaceEthn: Race/Ethnicity. 7 Levels (Derived from Hisp_Cat and Race_Cat2)

The RaceEthn was created with Hisp_Cat and Race_Cat2 variables. The RaceEthn has 7 categories: Hispanic; Non-Hispanic White; Non-Hispanic Black or African American; Non-Hispanic American Indian or Alaska Native; Non-Hispanic Asian; Non-Hispanic Native Hawaiian or other Pacific Islander; Non-Hispanic Multiple Races Mentioned. If Hisp_Cat had value of 10, "Not Hispanic", and Race_Cat2 had value of -9, "Missing data (Not Ascertained)", the RaceEthn was assigned with value of -9. The RaceEthn was assigned with value of -9 if Hispanic=-9.

RaceEthn5: Race/Ethnicity. 5 Levels (Derived from Hisp_Cat and Race_Cat2)

The RaceEthn5 was created with Hisp_Cat and Race_Cat2 variables. The RaceEthn5 has 5 categories: Non-Hispanic White; Non-Hispanic Black or African American; Hispanic; Non-Hispanic Asian; Non-Hispanic Other. If Hisp_Cat had value of 10, "Not Hispanic", and Race_Cat2 had value of -9, "Missing data (Not Ascertained)", the RaceEthn5 was assigned with value of -9. The RaceEthn5 was assigned with value of -9 if Hispanic=-9.

HHInc: What is your {combined} annual household income? 5 Levels (Derived from IncomeRanges)

The IncomeRanges variable was re-coded into 5 categories: Less than $20,000; $20,000 to < $35,000; $35,000 to < $50,000; $50,000 to < $75,000; $75,000 or more. The original negative values were carried over.

BMI: Body Mass Index (Weight*703)/(Height in inches**2)

The BMI variable was created with weight in pounds and height in inches. If feet of height or weight had value of -9, "Missing data (Not Ascertained)" but neither had value of -4, "Unreadable or Non-conforming numeric response", the BMI was assigned to -9. If feet of height or weight had value of -4, "Unreadable or Non-conforming numeric response", the BMI was assigned to -4.

AgeDX: At what age were you diagnosed with cancer? (Derived from WhenDiagnosedCancer)

The variable AgeDX is a copy of variable WhenDiagnosedCancer.

TimeSinceDX: How long ago were you diagnosed with cancer? (Derived from WhenDiagnosedCancer and Age)

The variable TimeSinceDX was created with EverHadCancer, WhenDiagnosedCancer and Age variables. The variable TimeSinceDX has 4 categories: Less than 1 Year since DX; 2-5 Years since DX; 6-10 Years since DX; 11+ Years since DX. If the variable EverHadCancer had value of 1 and either Age or

WhenDiagnosedCancer had value of -9, “Missing data (Not Ascertained)”, the TimeSinceDX was assigned to -9. If the variable EverHadCancer had value of 1 and WhenDiagnosedCancer is greater than Age, the TimeSinceDX was assigned to -4. If the variable EverHadCancer had value of -9, the TimeSinceDX was assigned to -6. If the variable EverHadCancer had value of 2 and WhenDiagnosedCancer had value of -2, the TimeSinceDX was assigned to -4. If the variable EverHadCancer had value of 2 and WhenDiagnosedCancer had value of -1, the TimeSinceDX was assigned to -1.

smokeStat: Smoking Status (Derived from Smoke100 and SmokeNow)

The variable smokeStat was created with Smoke100 and SmokeNow variables. The variable smokeStat has 3 categories: Current; Former; Never. If Smoke100 had value of 1 and SmokeNow had value of -5, “Multiple responses selected in error”, the smokeStat was assigned to -4. If Smoke100 had value of 1 and SmokeNow had value of -9, “Missing data (Not Ascertained)”, the smokeStat was assigned to -9. If Smoke100 had value of -9, “Missing data (Not Ascertained)”, the smokeStat was assigned to -6.

PHQ4: PHQ-4 total score (Derived composite from LittleInterest, Hopeless, Nervous, and Worrying)

The variable PHQ4 was created with LittleInterest, Hopeless, Nervous and Worrying variables. We created total score as continuous variable: 1) Rescore variables 0-3 and then reverse coding such that 'Not at all=0', 'Several Days'=1, 'More than half the days'=2, 'Nearly every day'=3 2) Compute total score by summing across 4 items 3) Total score range will be 0-12. If one of LittleInterest, Hopeless, Nervous and Worrying variables had value of -5, “Multiple response selected in error”, the PHQ was assigned to -5. If one of LittleInterest, Hopeless, Nervous and Worrying variables had value of -9, “Missing data (Not Ascertained)”, the PHQ was assigned to -9.

WeeklyMinutesModerateExercise: Minutes per week of at least moderate intensity exercise (Derived from TimesModerateExercise and HowLongModerateExerciseMinutes)

The variable WeeklyMinutesModerateExercise was created with TimesModerateExercise and HowLongModerateExerciseMinutes variables. If TimesModerateExercise is 0 then WeeklyMinutesModerateExercise was assigned to 0. If TimesModerateExercise is less than 0 then WeeklyMinutesModerateExercise was assigned to -9. If HowLongModerateExerciseMinutes is less than 0 then WeeklyMinutesModerateExercise was assigned to -9.

eCigUse: Electronic Cigarette Use (Derived from UsedECigEver and UseECigNow Recode)

The variable eCigUse was created with UsedECigEver and UseECigNow variables. The variable eCigUse has 3 categories: Current; Former; Never. If UsedECigEver had value of 1 and UseECigNow had value of -5, “Multiple responses selected in error”, the eCigUse was assigned to -4. If UsedECigEver had value of 1 and UseECigNow had value of -9, “Missing data (Not Ascertained)”, the eCigUse was assigned to -9. If UsedECigEver had value of -9, “Missing data (Not Ascertained)”, the eCigUse was assigned to -6.

AvgDrinksPerWeek: Average number of drinks per week (Derived from DrinkDaysPerWeek and DrinksPerDay)

The variable AvgDrinksPerWeek was created with DrinkDaysPerWeek and DrinksPerDay variables. If DrinkDaysPerWeek is less than 0 then AvgDrinksPerWeek was assigned to -9. If DrinksPerDay is less than 0 then AvgDrinksPerWeek was assigned to -9. The respondent with an outlying value of 40 for

DrinksPerDay (see Data Editing above) was assigned a value of -4 for AvgDrinksPerWeek. One additional respondent had an outlying value of 210 AvgDrinksPerWeek, which was recoded to -4.

PCCScale: -- > Patient Centered Communication scale

The PCCScale variable was created with ChanceAskQuestions, FeelingsAddressed, InvolvedDecisions, UnderstoodNextsteps, ExplainedClearly, SpentEnoughTime, and HelpUncertainty. We reverse the values of these 7 variables and take the mean of these 7 variables (if at least half the variables have valid values). And then we transform the mean value linearly to a 0-100 scale. If PCCScale value is missing, then we assigned value of -9 to it.

Label Editing

Labels Added for Standard Recode Variables

Labels were created for the following recoded variables:, AgeGrpA, AgeGrpB, EducA, EducB, RaceEthn, RaceEthn5, HHInc, BMI, AgeDX, TimeSinceDX, smokeStat, PHQ4, WeeklyMinutesModerateExercise, eCigUse, AvgDrinksPerWeek, PCCScale.

Labels Modified for Certain Variables

Labels were modified for the following variables to correct typos and improve clarity: Beer_CancerRisk, Beer_HeartRisk.

Format Editing

Formats Added for Standard Recode Variables

The formats AgeGrpA, AgeGrpB, EducA, EducB, RaceEthn, RaceEthn5f, HHInc, BMI, AgeDX, TimeSinceDX, smokeStat, phq4f, WeeklyMinutesModerateExercise, ECigStat, AvgDrinksPerWeek, and pccscale were created and assigned to the variables AgeGrpA, AgeGrpB, EducA, EducB, RaceEthn, RaceEthn5, HHInc, BMI, AgeDX, TimeSinceDX, smokeStat, PHQ4, WeeklyMinutesModerateExercise, eCigUse, AvgDrinksPerWeek, and PCCScale respectively.

Formats Modified for Certain Variables

All skip patterns in formats were modified (i.e. all instructions to skip questions were deleted). The modified formats were: ACCESSO, ADULTSI, CAOTHER, CAREGIH, CAREGIK, DRINKDA, EVERHAD, EVERHADC, FREQGOP, GENDERF, HADTEST, HAVEDEF, HAVEDEG, HEALTHF, HEARDGG, HEARDHP, INVITED, OCCUPAF, SEEKCAN, SEENFED, SEXUALF, SMOKE1F, TABLETH, TIMESMO, USEDECI, USEINTE, and WEARABL. On May 19, 2022, format USEDHEAL was modified to correct valid skip pattern. USEDHEAL -1 was modified from "Inapplicable, coded 2 in UseInternet" into "Inapplicable, coded 1 in HaveDevice_CellPh or HaveDevice_None, or coded 2 or 3 in TabletHealthWellnessApps".

All “&” symbols in formats were removed. The modified formats were: DMA.

The format DATE9 was applied to the UpdateDate variable to display the date values in dd-mmm-yyyy format.

Formats Removed for Dropped Variables

The format HIGHPAN was removed to reflect removal of variables HIGHSPANLI, HISPSURNAME, and HISP_HH. The format Qdisp was removed to reflect removal of variable QDISP.

The format FIPST was updated to $FIPST to account for the conversion from numeric to character to allow for a leading zero in the variable values.

Imputation of Income Variable

The income variable (IncomeRanges) has relatively higher percentage (11% for un-weighted percentage or 8% for weighted percentage) of missing values. This variable was imputed via PROC IMPUTE in SUDAAN. The imputation class variables are: Education (P7), RaceEthn (standard recode), and FullTimeOcc_Cat (derived variable from P4 and P5). The copy variables of the imputation class variables and income variable were created, where the missing values were appropriately coded. The copy variables are used for the imputation. The imputed values were saved in a new variable IncomeRanges_IMP.

SAS Code for Data Editing

```
* Recode MailHHAdults of 0 to 1 ;
if MailHHAdults=0 then MailHHAdults = 1;

* Recode Height_Feet for Anyone who reports height as 3 feet or less
or as 8 or more              *;
if 0<=Height_Feet<=3 or Height_Feet>=8 then Height_Feet = -4;

* Recode Height_Inches of -9 and -4 to 0;
if Height_Inches <0 then    Height_Inches = 0;

* Recode Weight less than or equal to 35 to Unreadable or Non-
conforming numeric response          ;
if 0<=Weight <=35 then  Weight = -4;


* Recode SelfAge between 0 and 17 to Unreadable or Non-conforming
numeric response                *;
if 0 <= SelfAge < 18 then  SelfAge = -4;

* Recode HHAdultAge2-HHAdultAge5 between 0 and 17 to Unreadable or
Non-conforming numeric response*;
array a(*) HHAdultAge2-HHAdultAge5;

do I = 1 to dim(a);
     if 0 <= a(I) < 18 then
     a(I) = -4;
end;

* Recode R_HHAdults of missing (.) to -9  ;
if missing(R_HHAdults) = 1 then R_HHAdults = -9;

* Recode Totalhousehold of 0 to 1;
if Totalhousehold=0 then Totalhousehold=1;
```

```
* Recode an outlier of DrinksPerDay (=50) to Unreadable or Non-
conforming numeric response        *;
     if DrinksPerDay=50 then DrinksPerDay=-4;



* Recode outliers of AverageTimeSitting (>20 hours)          *;
     if AverageTimeSitting > 20 then AverageTimeSitting = -4;
```

SAS Code for Standard Recode

```
     if 18 <= Age <= 34 then
          AgeGrpA = 1;
     else if 35 <= Age <= 39 then
          AgeGrpA = 2;
     else if 40 <= Age <= 44 then
          AgeGrpA = 3;
     else if 45 <= Age then
          AgeGrpA = 4;
     else if Age in (-1,-4, -9) then
          AgeGrpA = Age;
     label AgeGrpA = 'AgeGrpA. 4 Level Age Categories Version A
(Derived from Age; see History Document for more information)';
     if 18 <= Age <= 34 then
          AgeGrpB = 1;
     else if 35 <= Age <= 49 then
          AgeGrpB = 2;
     else if 50 <= Age <= 64 then
          AgeGrpB = 3;
     else if 65 <= Age <= 74 then
          AgeGrpB = 4;
     else if 75 <= Age then
          AgeGrpB = 5;
     else if Age in (-1,-4, -9) then
          AgeGrpB = Age;
     label AgeGrpB = 'AgeGrpB. 5 Level Age Categories Version B
(Derived from Age; see History Document for more information)';

     if Education in (1, 2) then
          EducA = 1;
     else if Education in (3) then
          EducA = 2;
     else if Education in (4, 5) then
          EducA = 3;
     else if Education in (6, 7) then
          EducA = 4;
     else if Education in (-1,-4,-9) then
          EducA = Education;
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      label EducA = 'EducA.  What is the highest level of school you
completed? 4 Levels (Derived from Education; see History Document for
more information)';

      if Education in (1, 2) then
            EducB = 1;
      else if Education in (3) then
            EducB = 2;
      else if Education in (4, 5) then
            EducB = 3;
      else if Education in (6) then
            EducB = 4;
      else if Education in (7) then
            EducB = 5;
      else if Education in (-1,-4,-9) then
            EducB = Education;
      label EducB = 'EducB.  What is the highest level of school you
completed? 5 Levels (Derived from Education; see History Document for
more information)';

      if Hisp_Cat in (21, 22, 23, 24, 25) then
            RaceEthn = 1;
      else if Hisp_Cat in (10) then
            do;
                  if Race_Cat2 in (11) then
                        RaceEthn = 2;
                  else if Race_Cat2 in (12) then
                        RaceEthn = 3;
                  else if Race_Cat2 in (14) then
                        RaceEthn = 4;
                  else if Race_Cat2 in (31, 32, 33, 34, 35, 36, 37) then
                        RaceEthn = 5;
                  else if Race_Cat2 in (51, 52, 53, 54) then
                        RaceEthn = 6;
                  else if Race_Cat2 in (16) then
                        RaceEthn = 7;
                  else if Race_Cat2 in (-1,-4,-9) then
                        RaceEthn = -9;
            end;
      else if Hisp_Cat in (-1,-4,-9) then
            do;
                  RaceEthn = -9;
            end;
      label RaceEthn = 'Race/Ethnicity. 7 Levels (Derived from Hisp_Cat
and Race_Cat2; see History Document for more information)';


      if Hisp_Cat in (21, 22, 23, 24, 25) then RaceEthn5 = 3;
      else if Hisp_Cat in (10) then do;
      if Race_Cat2 in (11) then RaceEthn5 = 1;
      else if Race_Cat2 in (12) then RaceEthn5 = 2;
```

```
      else if Race_Cat2 in (31, 32, 33, 34, 35, 36, 37) then RaceEthn5
      = 4;
      else if Race_Cat2 in (51, 52, 53, 54,14,16) then RaceEthn5 = 5;
      else if Race_Cat2 in (-1,-4,-9) then RaceEthn5 = -9;
      end;
      else if Hisp_Cat in (-1,-4,-9) then    do;
      RaceEthn5 = -9;
      end;
      label RaceEthn5 = 'Race/Ethnicity. 5 Levels (Derived from
Hisp_Cat and Race_Cat2; see History Document for more information)';

      if IncomeRanges in (1, 2, 3) then
            HHInc = 1;
      else if IncomeRanges in (4) then
            HHInc = 2;
      else if IncomeRanges in (5) then
            HHInc = 3;
      else if IncomeRanges in (6) then
            HHInc = 4;
      else if IncomeRanges in (7, 8, 9) then
            HHInc = 5;
      else if IncomeRanges in (-1,-4,-9) then
            HHInc = IncomeRanges;
      label HHInc = 'HHInc.  What is your {combined} annual household
income? 5 Levels (Derived from IncomeRanges; see History Document for
more information)';


      if Height_Feet >= 0 and Height_Inches >= 0 and Weight > 0 then
            BMI = (Weight * 703) / ((Height_Feet * 12 +
Height_Inches)**2);
      else if (Height_Feet in (-1, -9) and Weight >=-1 ) or
(Height_Feet >=-1 and Weight in (-1, -9)) or (Height_Feet = -9 and
Weight = -9) then
            BMI = -9;
      else if Height_Feet = -4 or Weight = -4 then
            BMI = -4;
      label BMI = 'BMI.  Body Mass Index (Weight*703)/(Height in
inches**2) (See History Document for more information)';
      if BMI not in (-4, -9) then
            BMI = round(BMI, 0.1);

      AgeDX = WhenDiagnosedCancer;
      AgeDX = 'AgeDX. At what age were you diagnosed with cancer?
(Derived from WhenDiagnosedCancer; see History Document for more
information)';

      if EverHadCancer in (1) then
            do;
                  if Age < 0 then
                        TimeSinceDX = Age;
                  else if WhenDiagnosedCancer <0 and Age >= 18 then
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```
                        TimeSinceDX = WhenDiagnosedCancer;
                  else if WhenDiagnosedCancer >= 0 and Age >= 18 then
                        do;
                              if 0 <= (Age - WhenDiagnosedCancer) <= 1
then
                                    TimeSinceDX = 1;
                              else if 2 <= (Age - WhenDiagnosedCancer) <=
5 then
                                    TimeSinceDX = 2;
                              else if 6 <= (Age - WhenDiagnosedCancer) <=
10 then
                                    TimeSinceDX = 3;
                              else if 11 <= (Age - WhenDiagnosedCancer)
then
                                    TimeSinceDX = 4;
                              else if (Age - WhenDiagnosedCancer) < 0
then
                                    TimeSinceDX = -4;
                        end;
            end;
      else if EverHadCancer in (-9) then
            TimeSinceDX = WhenDiagnosedCancer;
      else if EverHadCancer in (2) then
            do;
                  if WhenDiagnosedCancer in (-1) then
                        TimeSinceDX = WhenDiagnosedCancer;
                  else if WhenDiagnosedCancer in (-2) then
                        TimeSinceDX = -4;
            end;
      label TimeSinceDX = 'TimeSinceDX. How long ago were you diagnosed
with cancer? (Derived from WhenDiagnosedCancer and Age; see History
Document for more information)';

      if Smoke100 in (1) then
            do;
                  if SmokeNow in (1, 2) then
                        smokeStat = 1;
                  else if SmokeNow in (3) then
                        smokeStat = 2;
                  else if SmokeNow in (-5) then
                        smokeStat = -4;
                  else if SmokeNow in (-9) then
                        smokeStat = -9;
            end;
      else if Smoke100 in (2) then
            do;
                  smokeStat = 3;
            end;
      else if Smoke100 in (-9) then
            smokeStat = -6;
      label smokeStat = 'SmokeStat.  Smoking Status (Derived from
Smoke100 and SmokeNow; see History Document for more information)';
```

```
array b(*) LittleInterest Hopeless Nervous Worrying;
PHQ4= 0;
if b(1) = -5 or b(2) = -5 or b(3) = -5 or b(4) = -5 then PHQ4 =-5;
else do;
if b(1) = -9 or b(2) = -9 or b(3) = -9 or b(4) = -9 then PHQ4 = -9;
else do;
do I = 1 to dim(b);
if PHQ4 not in (-5, -9) and b(I) in (1, 2,3 ,4) then
PHQ4 = PHQ4 + (4-b(I));
end;
end;
end;
label PHQ4 = 'PHQ4. PHQ-4 total score (Derived composite from
LittleInterest, Hopeless, Nervous, and Worrying; see History Document
for more information)';
drop I;

if TimesModerateExercise=0 then WeeklyMinutesModerateExercise=0;
else if TimesModerateExercise<0 then WeeklyMinutesModerateExercise=-9;
else if HowLongModerateExerciseMinutes<0 then
WeeklyMinutesModerateExercise=-9;
else iF TimesModerateExercise>0 then do;
      if HowLongModerateExerciseMinutes = 0 then
WeeklyMinutesModerateExercise = 0;
      else WeeklyMinutesModerateExercise =
HowLongModerateExerciseMinutes*TimesModerateExercise;
end;
label WeeklyMinutesModerateExercise="WeeklyMinutesModerateExercise.
Minutes per week of at least moderate intensity exercise (Derived from
TimesModerateExercise and HowLongModerateExerciseMinutes; see History
Document for more information)";

if UsedECigEver in (1) then
      do;
            if UseECigNow in (1, 2) then
                  eCigUse = 1;
            else if UseECigNow in (3) then
                  eCigUse = 2;
            else if UseECigNow in (-5) then
                  eCigUse = -4;
            else if UseECigNow in (-9) then
                  eCigUse = -9;
      end;
else if UsedECigEver in (2) then
      do;
            eCigUse = 3;
      end;
else if UsedECigEver in (-9) then
      eCigUse = -6;
```

```
label eCigUse = 'eCigUse.  Electronic Cigarette Use (Derived from
UsedECigEver and UseECigNow; see History Document for more
information)';

if DrinkDaysPerWeek=0 then AvgDrinksPerWeek=0;
else if DrinkDaysPerWeek < 0 then AvgDrinksPerWeek=-9;
else if DrinksPerDay < 0 then AvgDrinksPerWeek=-9;
else if DrinkDaysPerWeek>0 then do;
     if DrinksPerDay=0 then AvgDrinksPerWeek = 0;
     else AvgDrinksPerWeek = DrinkDaysPerWeek*DrinksPerDay;
end;
*Recode outliers;
if AvgDrinksPerWeek = 210 then AvgDrinksPerWeek = -4;
label AvgDrinksPerWeek = "AvgDrinksPerWeek. Average number of
drinks per week (Derived from DrinkDaysPerWeek and DrinksPerDay;
see History Document for more information)";

if ChanceAskQuestions < 0 then ChanceAskQuestionsRev=.;
     if ChanceAskQuestions = 1 then ChanceAskQuestionsRev=4;
     if ChanceAskQuestions = 2 then ChanceAskQuestionsRev=3;
     if ChanceAskQuestions = 3 then ChanceAskQuestionsRev=2;
     if ChanceAskQuestions = 4 then ChanceAskQuestionsRev=1;
     ****************************;
     if FeelingsAddressed < 0 then FeelingsAddressedRev=.;
     if FeelingsAddressed = 1 then FeelingsAddressedRev=4;
     if FeelingsAddressed = 2 then FeelingsAddressedRev=3;
     if FeelingsAddressed = 3 then FeelingsAddressedRev=2;
     if FeelingsAddressed = 4 then FeelingsAddressedRev=1;
     ****************************;
     if InvolvedDecisions < 0 then InvolvedDecisionsRev=.;
     if InvolvedDecisions = 1 then InvolvedDecisionsRev=4;
     if InvolvedDecisions = 2 then InvolvedDecisionsRev=3;
     if InvolvedDecisions = 3 then InvolvedDecisionsRev=2;
     if InvolvedDecisions = 4 then InvolvedDecisionsRev=1;
     ****************************;
     if UnderstoodNextSteps < 0 then UnderstoodNextStepsRev=.;
     if UnderstoodNextSteps = 1 then UnderstoodNextStepsRev=4;
     if UnderstoodNextSteps = 2 then UnderstoodNextStepsRev=3;
     if UnderstoodNextSteps = 3 then UnderstoodNextStepsRev=2;
     if UnderstoodNextSteps = 4 then UnderstoodNextStepsRev=1;
     ****************************;
     if ExplainedClearly < 0 then ExplainedClearlyRev=.;
     if ExplainedClearly = 1 then ExplainedClearlyRev=4;
     if ExplainedClearly = 2 then ExplainedClearlyRev=3;
     if ExplainedClearly = 3 then ExplainedClearlyRev=2;
     if ExplainedClearly = 4 then ExplainedClearlyRev=1;
     ****************************;
     if SpentEnoughTime < 0 then SpentEnoughTimeRev=.;
     if SpentEnoughTime = 1 then SpentEnoughTimeRev=4;
     if SpentEnoughTime = 2 then SpentEnoughTimeRev=3;
     if SpentEnoughTime = 3 then SpentEnoughTimeRev=2;
     if SpentEnoughTime = 4 then SpentEnoughTimeRev=1;
     ****************************;
     if HelpUncertainty < 0 then HelpUncertaintyRev=.;
     if HelpUncertainty = 1 then HelpUncertaintyRev=4;
```

```
      if HelpUncertainty = 2 then HelpUncertaintyRev=3;
      if HelpUncertainty = 3 then HelpUncertaintyRev=2;
      if HelpUncertainty = 4 then HelpUncertaintyRev=1;
      ***************************;
** create the PCCMean variable;
If nmiss
(ChanceAskQuestionsRev,FeelingsAddressedRev,InvolvedDecisionsRev,UnderstoodNe
xtStepsRev,ExplainedClearlyRev,SpentEnoughTimeRev,HelpUncertaintyRev)<4
Then PCCMean =
MEAN(ChanceAskQuestionsRev,FeelingsAddressedRev,InvolvedDecisionsRev,Understo
odNextStepsRev,ExplainedClearlyRev,SpentEnoughTimeRev,HelpUncertaintyRev);

/*Creat composite score - linear variable on a scale of 100*/
PCCScale=round(((pccmean-1)*100)/3,0.1);

      if PCCScale < 0 then PCCScale=.;
label PCCScale='PCCScale. Patient Centered Communication scale';
```

SAS Code for Format Editing

SAS Code for Formats Added for Standard Recode Variables

```
value AgeGrpA
1 = '18-34'
2 = '35-39'
3 = '40-44'
4 = '45+'
-4 = 'Unreadable or Nonconforming Numeric Response'
-9 = 'Missing Data (Not Ascertained)'
;

value AgeGrpB
1 = '18-34'
2 = '35-49'
3 = '50-64'
4 = '65-74'
5 = '75+'
-4 = 'Unreadable or Nonconforming Numeric Response'
-9 = 'Missing Data (Not Ascertained)'
;

value EducA
1 = 'Less than High School'
2 = 'High School Graduate'
3 = 'Some College'
4 = 'College Graduate or More'
-9 = 'Missing Data (Not Ascertained)'
;

value EducB
1 = 'Less than High School'
2 = 'High School Graduate'
```

```
3 = 'Some College'
4 = "Bachelor's Degree"
5 = 'Post-Baccalaureate Degree'
-9 = 'Missing Data (Not Ascertained)'
;

value RaceEthn
1 = 'Hispanic'
2 = 'Non-Hispanic White'
3 = 'Non-Hispanic Black or African American'
4 = 'Non-Hispanic American Indian or Alaska Native'
5 = 'Non-Hispanic Asian'
6 = 'Non-Hispanic Native Hawaiian or other Pacific Islander'
7 = 'Non-Hispanic Multiple Races Mentioned'
-4 = 'Unreadable or Nonconforming Numeric Response'
-9 = 'Missing Data (Not Ascertained)'
;

value RaceEthn5f
1 = "Non-Hispanic White"
2= "Non-Hispanic Black or African American"
3="Hispanic"
4="Non-Hispanic Asian"
5="Non-Hispanic Other"
-9="Missing Data--Not Ascertained"
;

value HHInc
1 = 'Less than $20,000'
2 = '$20,000 to < $35,000'
3 = '$35,000 to < $50,000'
4 = '$50,000 to < $75,000'
5 = '$75,000 or More'
-5 = 'Multiple Responses Selected in Error'
-9 = 'Missing Data (Not Ascertained)'
;

value BMI
-4 = 'Unreadable or Nonconforming Numeric Response'
-9 = 'Missing Data (Not Ascertained)'
;

value AgeDX
-1 = 'Inapplicable, coded 2 in EverHadCancer'
-2 = 'Question Answered in Error (Commission Error)'
-6 = 'Missing Data (Filter Missing)'
-9 = 'Missing Data (Not Ascertained)'
;

value TimeDX
1 = 'Less than 1 Yr Since DX'
2 = '2-5 Yrs Since DX'
```

```
3 = '6-10 Yrs Since DX'
4 = '11+ Yrs Since DX'
-1 = 'Inapplicable, coded 2 in EverHadCancer'
-4 = 'Unreadable or Nonconforming Numeric Response'
-6 = 'Missing Data (Filter Missing), coded -9 in EverHadCancer'
-9 = 'Missing Data (Not Ascertained)'
;

value smStat
1 = 'Current'
2 = 'Former'
3 = 'Never'
-4 = 'Unreadable or Nonconforming Numeric Response'
-6 = 'Missing Data (Filter Missing), coded -9 in Smoke100'
-9 = 'Missing Data (Not Ascertained)'
;

value phq4f
-5 = 'Multiple Responses Selected in Error'
-9 = 'Missing Data (Not Ascertained)'
;

value WeExer
-5 = 'Multiple Responses Selected in Error'
-9 = 'Missing Data (Not Ascertained)'
;

value ecigStat
1 = 'Current'
2 = 'Former'
3 = 'Never'
-4 = 'Unreadable or Nonconforming Numeric Response'
-6 = 'Missing Data (Filter Missing), coded -9 in UsedECigEver'
-9 = 'Missing Data (Not Ascertained)'
;

value AvgDri
-4 = 'Unreadable or Nonconforming Numeric Response'
-5 = 'Multiple Responses Selected in Error'
-9 = 'Missing Data (Not Ascertained)'
;

value pccscale
-9 = 'Missing data (Not Ascertained)'
;

format     AgeGrpA AgeGrpA.
           AgeGrpB AgeGrpB.
           EducA EducA.
           EducB EducB.
           RaceEthn RaceEthn.
           RaceEthn5 RaceEthn5f.
           HHInc HHInc.
```

```
                BMI BMI.
                AgeDX AgeDX.
                TimeSinceDX TimeDX.
                smokeStat smStat.
                phq4 phq4f.
                WeeklyMinutesModerateExercise  WeExer.
                ECigUse ECigStat.
                IncomeRanges_IMP  incomer.
                AvgDrinksPerWeek AvgDri.
                PCCScale pccscale. ;
```

SAS Code for Imputation of Income Variable

```
* Impute IncomeRanges via PROC HOTDECK
*;
data HINTS5CYCLE4;
      set HINTS5CYCLE4;

      COPY_Education = Education;
      if COPY_Education in (-9) then
            COPY_Education = .;

      COPY_RaceEthn = RaceEthn;
      if COPY_RaceEthn in (-9) then
            COPY_RaceEthn = .;

  COPY_FullTimeOcc_Cat = FullTimeOcc_Cat;
      if FullTimeOcc_Cat in (-4, -9) then
            COPY_FullTimeOcc_Cat = .;


      COPY_IncomeRanges = IncomeRanges;
      if COPY_IncomeRanges in (-9) then
            COPY_IncomeRanges = .;

      ID = _N_;

      format COPY_Education Educati. COPY_RaceEthn RaceEthn.
COPY_FullTimeOcc_Cat FullTim. COPY_IncomeRanges IncomeR.

run;

proc freq data= HINTS5CYCLE4;
      tables COPY_Education*Education / list missing;
      tables COPY_RaceEthn*RaceEthn / list missing;
      tables COPY_FullTimeOcc_Cat*FullTimeOcc_Cat/ list missing;
      tables COPY_IncomeRanges*IncomeRanges / list missing;
run;


proc impute data= HINTS5CYCLE4 method=wshd notsorted;
      weight person_finwt0;
```

```
      impvar COPY_IncomeRanges;
      impby COPY_Education COPY_RaceEthn COPY_FullTimeOcc_Cat;
      impname COPY_IncomeRanges="IncomeRanges_IMP";
      impid ID;
      output IMPID IMPBY IMPUTEVAL / filename=impute1 replace;
run;

proc freq data=impute1;
      tables IncomeRanges_IMP / missing;
run;

proc contents data=impute1;
run;

proc sort data= HINTS5CYCLE4;
      by ID;
run;

proc sort data=impute1 (keep=ID IncomeRanges_IMP);
      by ID;
run;

data HINTS5CYCLE4;
      merge HINTS5CYCLE4 (in=A) impute1 (in=B);
      by ID;

      if A = 1 and B = 1;
run;

data _null_;
      set HINTS5CYCLE4;

      if IncomeRanges not in (-9) and COPY_IncomeRanges ^=
      IncomeRanges_IMP then
            put ID IncomeRanges COPY_IncomeRanges IncomeRanges_IMP;
run;

data HINTS5CYCLE4;
      set HINTS5CYCLE4;

      if missing(IncomeRanges_IMP) = 1 then
            IncomeRanges_IMP = IncomeRanges;
      label IncomeRanges_IMP = '-->IncomeRanges_IMP.  Imputed
      IncomeRanges variable via PROC HOTDECK in SUDAAN (see History
      Document for more information) ';
      format IncomeRanges_IMP IncomeR.;

      drop COPY_Education COPY_RaceEthn COPY_FullTimeOcc_Cat
            ID
            COPY_IncomeRanges;
run;
```